Semantics: studying meanings and functions of words

Lexicology and Lexicography — ***Course Website***

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Outline

- **Discussion of study questions** — Q1–Q7 on Hilpert, Saavedra, and Rains (2023)
- **Theoretical framework** — Quick review: Principle of No Synonymy, Distributional Hypothesis
- **COCA study overview** — Data, method, and results
- **Practice using Sketch Engine** — Analyse clipping pairs using collocations and word sketches

Preparatory reading: Hilpert, Saavedra, and Rains (2023)

Study questions

Please read the paper and prepare answers to these questions for our class discussion:

1. What is the Principle of No Synonymy, and what does it predict about clippings and their source words?
2. How do the authors use the principle "You shall know a word by the company it keeps" (**Firth 1957**) to study meaning differences? What do they compare?
3. What does the paper find about clippings in "involved" vs. "informational" contexts? Which linguistic features distinguish them?
4. Some clippings (e.g., *fridge/refrigerator*) are nearly synonymous, while others (e.g., *cardio/cardiovascular*) show clear differences. What factors might explain this variation?
5. How can collocation analysis reveal semantic differences not obvious from intuition? Use an example from the paper.
6. Choose one pair from Table 1 (e.g., *exam/examination*). What differences might you expect, and what would you look for in a corpus analysis?
7. The paper offers two explanations for clippings in "involved" contexts: familiarity vs. efficiency. What evidence supports each? Which do you find more convincing?

Q1: Principle of No Synonymy

What is the Principle of No Synonymy, and what does it predict about clippings and their source words?

- **Definition**: A difference in linguistic form always indicates a difference in meaning.
- **Broad notion of "meaning"**: semantic content, but also discourse-functional, interpersonal, and information-structural aspects.
- **Prediction for clippings**: clippings and source words should have distinct distributional characteristics that reflect functional differences.

Example: *brother* (kinship term) vs *bro* (address term for close friend within specific communities) — clearly distinct meanings despite shared etymology.

Q2: Distributional Hypothesis

How do the authors use the principle “You shall know a word by the company it keeps” to study meaning differences? What do they compare?

Two levels of comparison:

1. **Aggregate level**: frequency of “involved” vs “informational” markers across all 50 pairs (Table 2)
2. **Individual level**: token-based semantic vector spaces using second-order collocates

Q3: Involved vs Informational Contexts

What does the paper find about clippings in "involved" vs "informational" contexts? Which linguistic features distinguish them?

The paper tests 12 markers of involved vs informational text production.

Involved markers (more frequent with clippings):

- Private verbs (*know*, *think*, *believe*)
- Contractions (*it's*, *you're*)
- Present tense verbs
- 1st and 2nd person pronouns (*I*, *you*)
- Indefinite pronouns (*something*, *anyone*)
- Pronoun *it*

Informational markers (more frequent with source words):

- Complementiser *that*
- Common nouns
- Prepositions

Q4: Variation in Synonymy

Some clippings (e.g., *fridge/refrigerator*) are nearly synonymous, while others (e.g., *cardio/cardiovascular*) show clear differences. What factors might explain this variation?

Clearly distinct pairs (Figure 3):

- *cardio/cardiovascular*: fitness vs medical contexts
- *chemo/chemotherapy*: patient experience vs technical medical
- *indie/independent*: underground culture vs general meaning
- *dorm/dormitory*: includes tourist lodging uses

Near-synonymous pairs (Figure 4):

- *fridge/refrigerator*: same semantic facets accessible to both
- *porn/pornography*: industry, psychological, legal aspects shared

Q5: Collocation Analysis

How can collocation analysis reveal semantic differences not obvious from intuition? Use an example from the paper.

Example: *cardio* vs *cardiovascular*

Intuition might suggest they mean the same thing. But collocates reveal:

cardiovascular:

- *disease*, *diabetes*, *chronic*
- *renal*, *kidney*, *liver*
- Medical/clinical contexts

cardio:

- *workout*, *fat-blasting*, *calendar*
- *recipes*, *oranges*, *dishes*
- Fitness/lifestyle contexts

Q6: Corpus Analysis of *exam/examination*

Choose one pair from Table 1 (e.g., *exam/examination*). What differences might you expect, and what would you look for in a corpus analysis?

Expected differences:

- *exam*: academic/medical test contexts, informal register
- *examination*: formal register, plus “investigation/scrutiny” sense

What to look for in a corpus:

- **Collocates**: *final exam*, *take an exam* vs *detailed examination*, *close examination*, *under examination*
- **Text types**: spoken/informal (blog, fiction) vs written/formal (academic, news)
- **Modifiers**: different adjectives? (*tough exam* vs *thorough examination*)
- **Syntactic patterns**: *examination of* (investigation) vs *exam in/on* (subject)?

Q7: Familiarity vs Efficiency

The paper offers two explanations for clippings in “involved” contexts: familiarity vs efficiency. What evidence supports each? Which do you find more convincing?

Familiarity explanation:

- Clippings signal common ground, shared expertise, in-group membership
- Evidence: co-occurrence with “involved” markers; personal experience contexts
- Example: *chemo* in patient narratives vs *chemotherapy* in medical papers

Efficiency explanation:

- Shorter forms preferred when predictable; saves processing effort
- Counter-evidence: predictability effects are limited when forms differ in meaning

Theoretical framework

Does clipping involve a change in meaning?

One view: clippings and source words are alternatives that only differ in form, not meaning.

Examples:

- *mic* vs. *microphone*
- *lab* vs. *laboratory*
- *ad* vs. *advertisement*

But: do *brother* and *bro* have the same meaning?

The Principle of No Synonymy

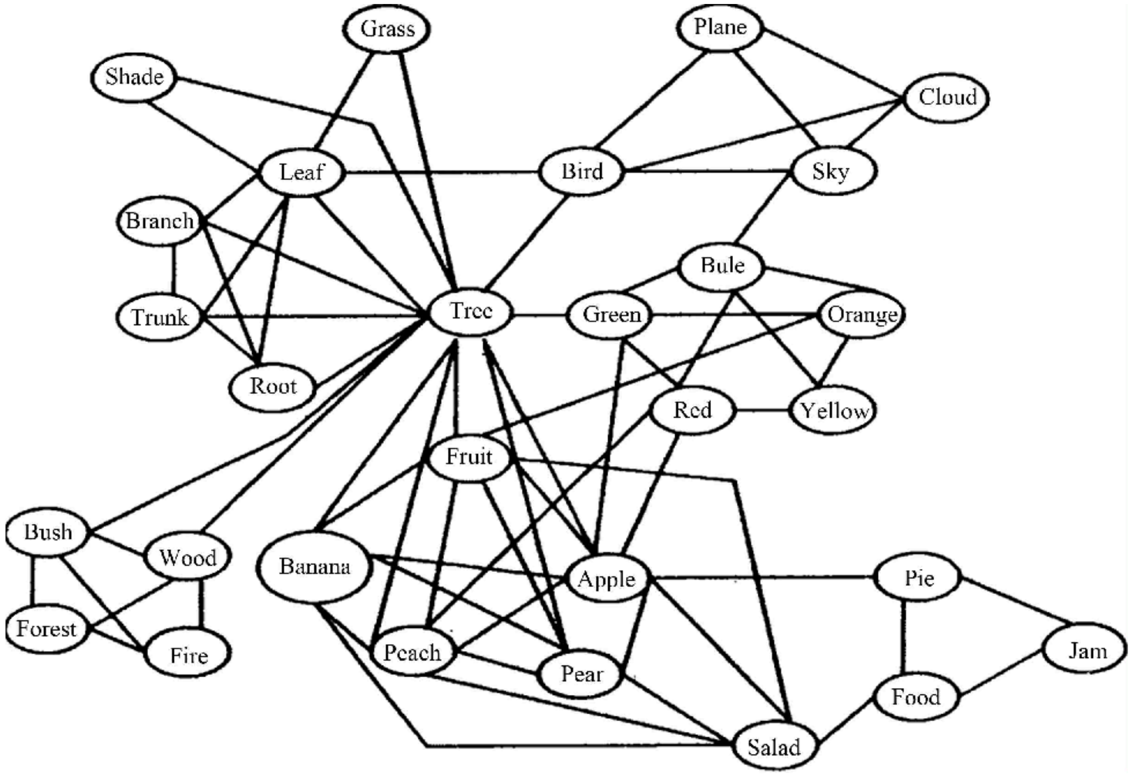
- A difference in linguistic form always indicates a difference in meaning (Goldberg 1995).
- "Meaning" is broad: semantic, discourse-functional, interpersonal, etc.
- **Prediction**: clippings and source words should have distinct distributions, reflecting functional differences.

> **Note**
>
> Think of 3 sentences the words *brother* and *bro* could appear in.

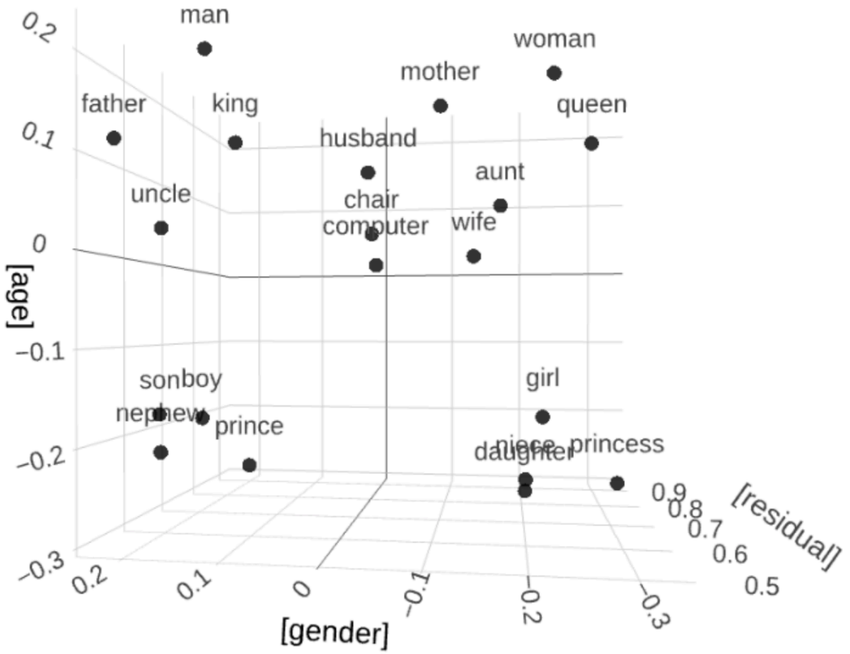
The Distributional hypothesis

- The meaning of words is reflected in their contextual elements in language use: **“You shall know a word by the company it keeps”** (Firth 1957).
- Words that appear in similar contexts share aspects of their meanings.
- Example: *happy* and *joyful* share collocates like *very*, *feel*, *make*, *person*.

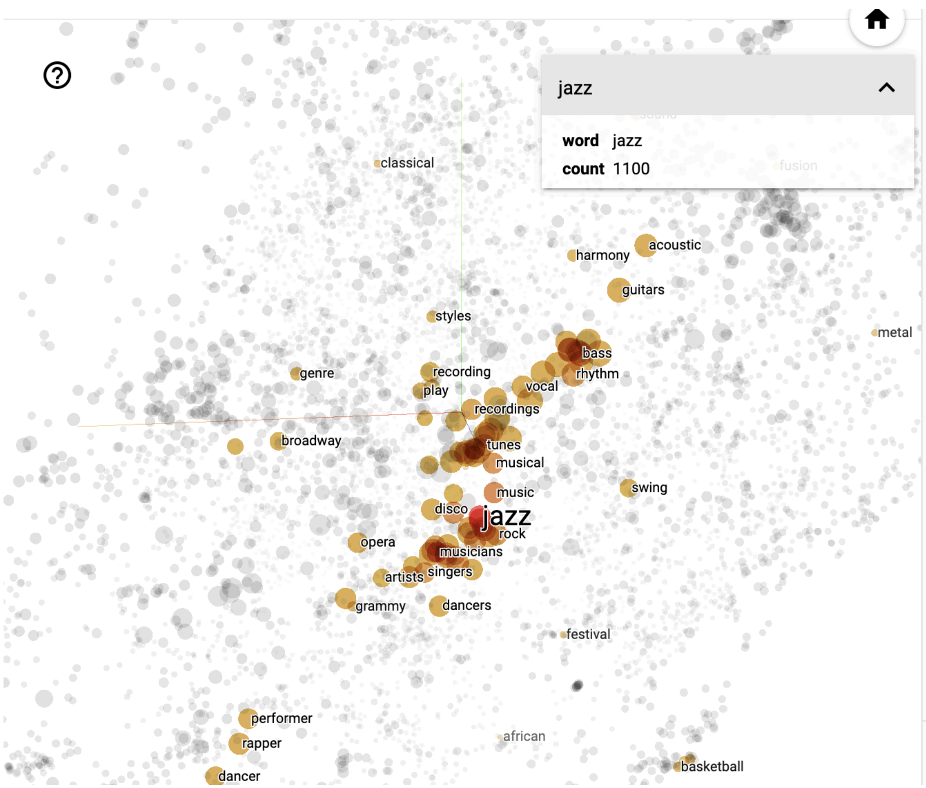
Cognitive-linguistic models of the lexicon



Word embeddings (Bandyopadhyay et al. 2022)



TensorFlow Projector



Exercise

Word Embeddings Explorer

1. Explore synonyms and antonyms (Similar words)
   - Enter a seed word (e.g., `happy_ADJ`).
   - Inspect the 5 nearest neighbours.
   - Compare with an antonym (e.g., `sad_ADJ`).
2. Visualise semantic clusters (Visualization)
   - Choose two related groups (e.g., animals vs furniture).
   - Plot in vector space; describe visible groupings.
3. Word analogies (Calculator)
   - Try analogies, e.g., `male_ADJ : doctor_NOUN = female_ADJ : ?`.
   - Test several analogies; note successes and failures.
4. Investigate ambiguity
   - Pick a polysemous word (e.g., *bank*).
   - Analyse how neighbours reflect different senses.

Studying the meaning of clippings in the COCA

Hilpert, Saavedra, and Rains (2023)

Data

- Sample of 50 English clippings and their corresponding source words.
- Corpus data from COCA (Davies 2008) for each clipping and source word.

Table 1. Selected clippings and their long forms.

	clipping	source word		clipping	source word		clipping	source word
1	admin	administration	18	expat	expatriate	35	mic	microphone
2	alum	alumnus	19	fab	fabulous	36	pic	picture
3	ammo	ammunition	20	fave	favorite	37	porn	pornography
4	amp	amplifier	21	frat	fraternity	38	prefab	prefabricated
5	app	application	22	fridge	refrigerator	39	prenup	prenuptial agreement
6	boomer	baby-boomer	23	gent	gentleman	40	prof	professor
7	burbs	suburbs	24	glam	glamorous	41	psych	psychology
8	carbs	carbohydrates	25	goth	gothic	42	pup	puppy
9	cardio	cardiovascular	26	hippo	hippopotamus	43	quake	earthquake
10	celeb	celebrity	27	improv	improvisation	44	rehab	rehabilitation
11	chemo	chemotherapy	28	indie	independent	45	roach	cockroach
12	chimp	chimpanzee	29	intro	introduction	46	stats	statistics
13	croc	crocodile	30	lab	laboratory	47	teen	teenager
14	decaf	decaffeinated	31	legit	legitimate	48	tux	tuxedo
15	dorm	dormitory	32	limo	limousine	49	undergrad	undergraduate
16	exam	examination	33	lube	lubricant	50	vid	video
17	exec	executive	34	memo	memorandum			

Method

Collocation analysis to investigate semantic profiles across contexts.

Table 4. Second-order collocates of tokens of *cardio* and *cardiovascular*.

Concordance line	Key word	Second-order collocates
(1a)	cardio	muscles, slam, abdominal, eyelids, stamina, circulation, nerves, headache, calves, ...
(1b)	cardio	recipes, oranges, cooked, pies, dishes, salads, sweets, foods, fried, dried, fruit, ...
(1c)	cardiovascular	diabetes, inflammatory, disorders, chronic, disease, renal, arthritis, kidney, liver, ...
(1d)	cardiovascular	chronic, obstruction, lung, asthma, epithelial, disease, hypertension, prevalence, intermittent, ...

Results

Variation across text types

Table 2. Twelve features of involved and informational text production (Biber 1988: 102).

Involved Text Production	**Informational Text Production**
private verbs*	complementizer THAT
contractions	common nouns
present tense verbs	prepositions
1st person pronouns	type/token ratio
2nd person pronouns	
indefinite pronouns**	
pronoun IT	
demonstrative pronouns	

*Following Biber (1988: 242), the verbs that were included in this category were *anticipate*, *assume*, *believe*, *conclude*, *decide*, *demonstrate*, *determine*, *discover*, *doubt*, *estimate*, *fear*, *feel*, *find*, *forget*, *guess*, *hear*, *hope*, *imagine*, *imply*, *indicate*, *infer*, *know*, *learn*, *mean*, *notice*, *prove*, *realize*, *recognize*, *remember*, *reveal*, *see*, *show*, *suppose*, *think*, and *understand*.
**Following Biber (1988: 226), the pronouns that were included in this category were *anybody*, *anyone*, *anything*, *everybody*, *everyone*, *everything*, *nobody*, *none*, *nothing*, *nowhere*, *somebody*, *someone*, and *something*.

Differences in collocations for *cardiovascular* vs *cardio*

Table 4. Second-order collocates of tokens of *cardio* and *cardiovascular*.

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(1a)	cardio	muscles, slam, abdominal, eyelids, stamina, circulation, nerves, headache, calves, ...
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(1c)	cardiovascular	diabetes, inflammatory, disorders, chronic, disease, renal, arthritis, kidney, liver, ...
(1d)	cardiovascular	chronic, obstruction, lung, asthma, epithelial, disease, hypertension, prevalence, intermittent, ...

Practice: studying meanings using Sketch Engine

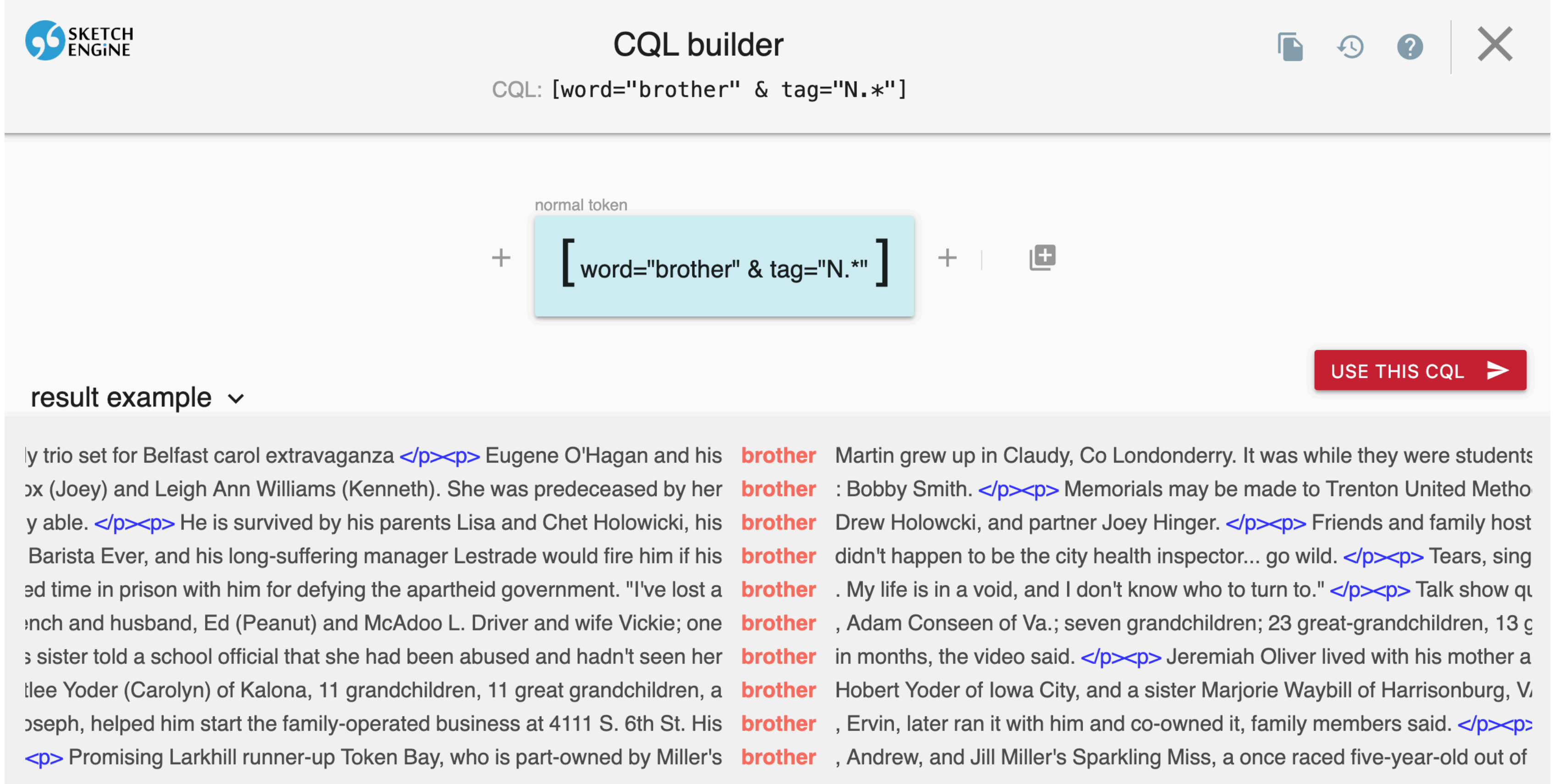
Your Task

1. Choose one of the word pairs from **the sample above** (**Hilpert, Saavedra, and Rains 2023**).
2. Use the collocations and word sketches in the **enTenTen21** corpus to analyse their meanings.
3. Use the **Guiding Questions** below to structure your analysis.
4. Take notes in **this PowerPoint file**.
5. Share your findings with the class.

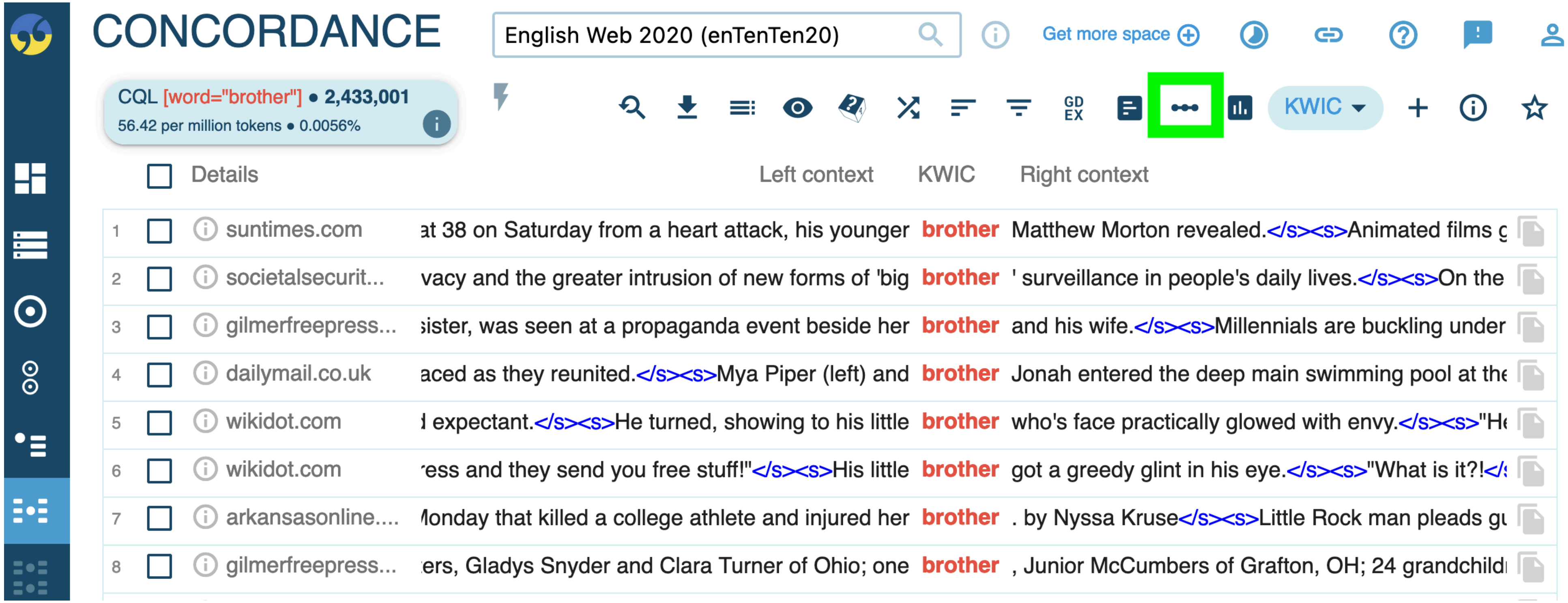
Collocation analysis

1. Run a query

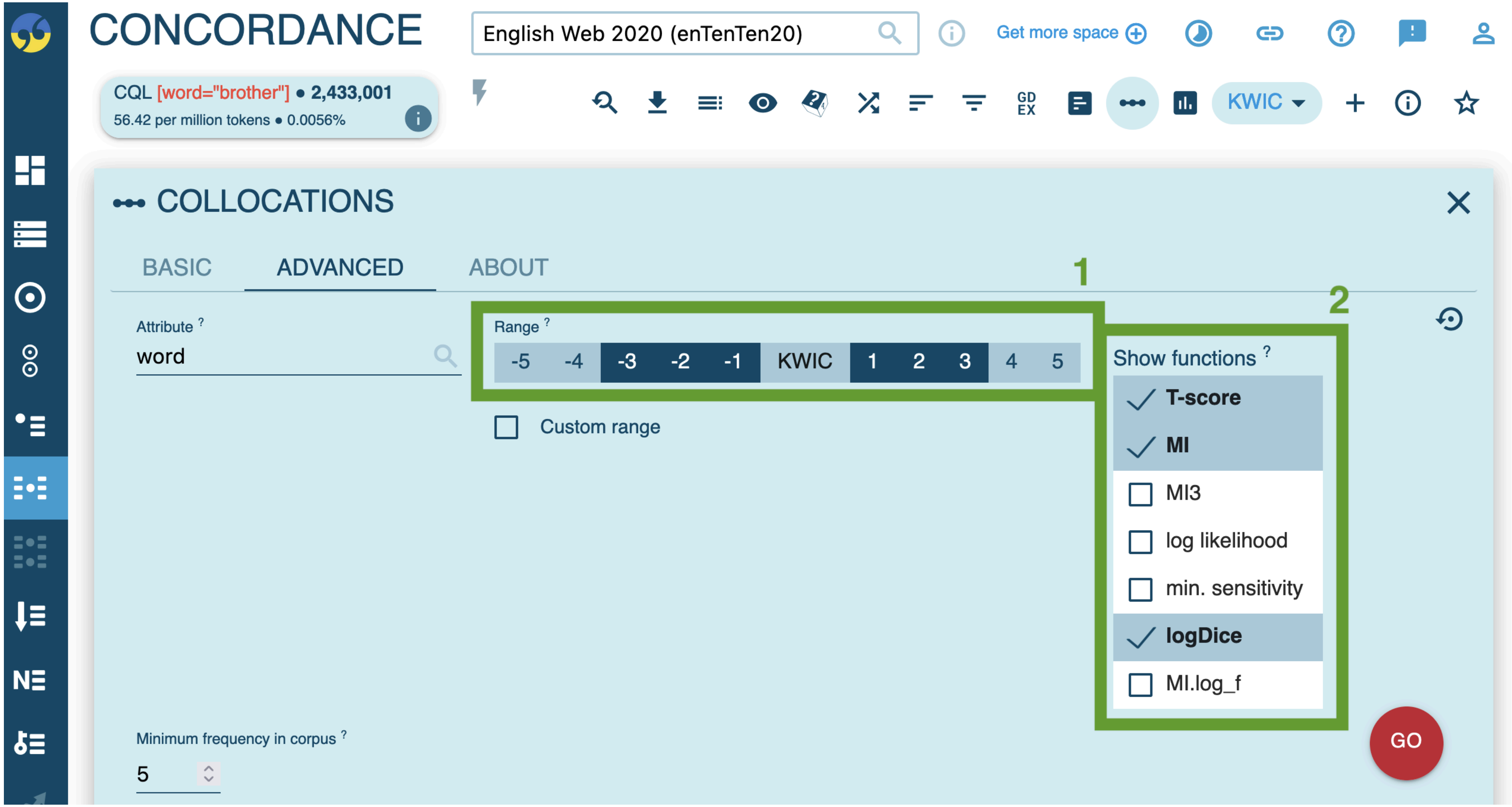
e.g. `[word="brother" & tag="N.*"]`



2. Click the collocation analysis icon



3. Configure window range and statistics



4. View the results and test collocation measures

CONCORDANCE

English Web 2020 (enTenTen20)

Get more space

CQL [word="brother"] • 2,433,001
56.42 per million tokens • 0.0056%

KWIC

Collocations

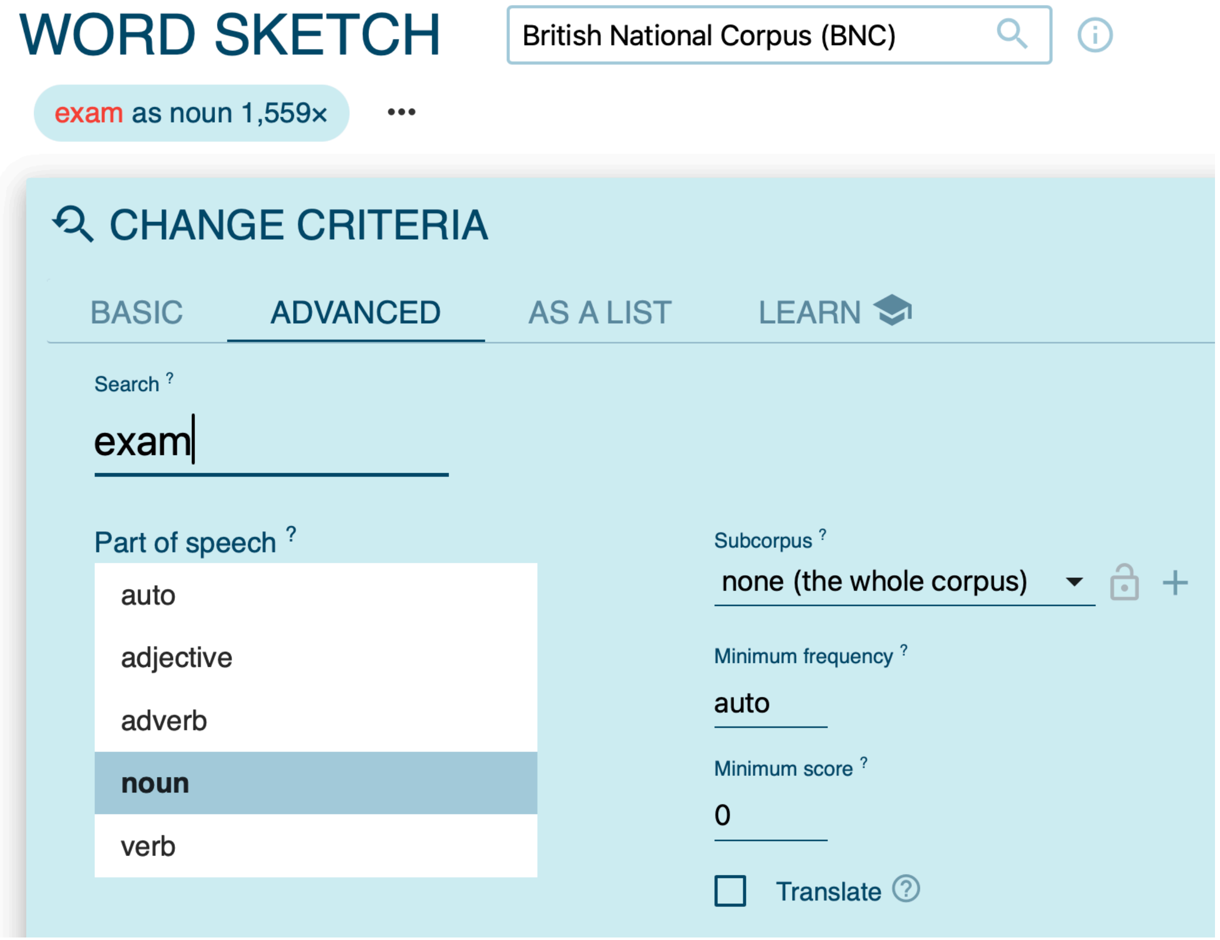
CHANGE CRITERIA | BACK TO CONCORDANCE

1

		Word	Cooccurrences ?	Candidates ?	T-score	MI	LogDice ↓	
1	☐	younger	142,791	1,474,707	377.66	10.75	10.23	•••
2	☐	older	111,838	2,800,488	333.95	9.47	9.45	•••
3	☐	sister	76,686	1,721,760	276.57	9.62	9.24	•••
4	☐	elder	44,609	301,639	211.13	11.36	9.06	•••
5	☐	twin	30,902	531,632	175.62	10.01	8.42	•••
6	☐	his	695,214	99,786,308	827.04	6.95	7.80	•••
7	☐	My	81,924	9,785,966	284.29	7.21	7.78	•••
8	☐	His	67,572	9,307,687	257.93	7.01	7.56	•••
9	☐	my	273,056	57,070,941	516.39	6.41	7.23	•••

Word sketches: Single forms

1. Generate a word sketch for a clipped form (e.g. exam), specifying the word class.



2. Examine syntactic contexts.

WORD SKETCH

English Web 2021 (enTenTen21)

Get more space

examination as noun 1,769,654×

modifiers of "examination"	
entrance entrance examination	8.6
thorough a thorough examination of	8.5
careful careful examination of the	8.3
cross cross examination	8.3
close closer examination	8.2
physical physical examination	7.9
microscopic microscopic examination of	7.7
forensic forensic examination	7.6

nouns modified by "examination"	
malpractice examination malpractice	7.1
finding physical examination findings	5.9
timetable examination timetable	5.8
glove examination gloves	5.7
hall the examination hall	5.5
fee examination fee	5.5
paper examination papers	5.4
syllabus examination syllabus	5.3

verbs with "examination" as object	
pass passed the examination	7.9
qualify qualifying examination	7.8
undergo undergo a medical examination	7.8
conduct examination conducted	7.6
administer examination administered	7.1
sit sit the examination	6.5
perform examination was performed	6.5
fail failed the examination	6.2

WORD SKETCH

English Web 2021 (enTenTen21)

Get more space

exam as noun 2,131,195×

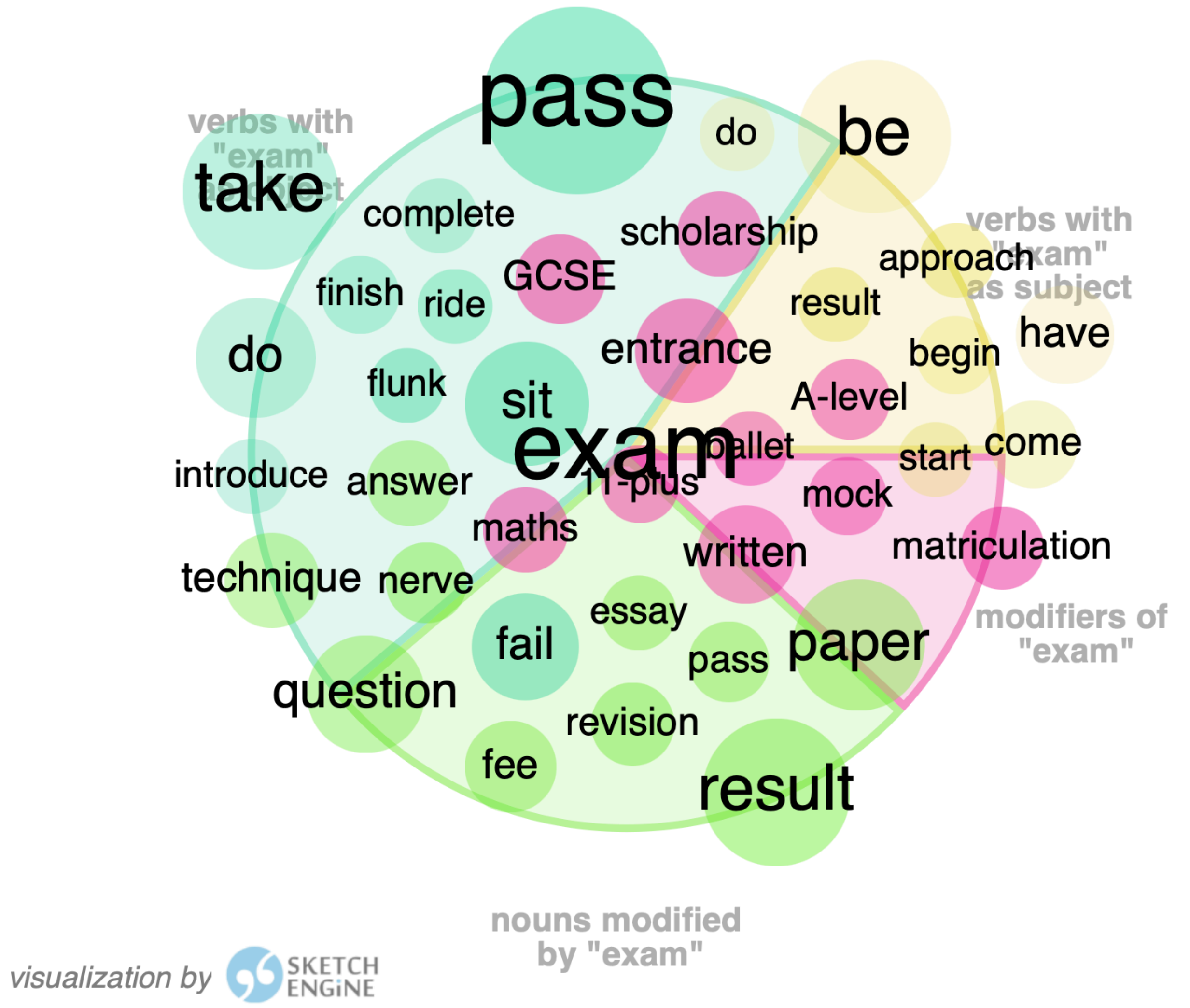
modifiers of "exam"		
certification	9.8	certification exam
entrance	9.4	entrance exam
practice	7.9	practice exams
final	7.7	final exam
eye	7.7	eye exam
AP	7.6	AP exams
bar	7.5	the bar exam
written	7.5	written exam

nouns modified by "exam"		
dump	9.0	exam dumps
syllabus	8.4	exam syllabus . Pass4sure
preparation	8.3	exam preparation
braindump	8.2	exam braindump
prep	8.1	exam prep
question	7.5	exam questions
Dumps	7.2	Real Certification Exam Dumps is Getting an
engine	7.0	interactive exam engine

verbs with "exam" as object		
pass	9.2	pass the exam
sit	8.2	sit the exam
fail	8.1	fail the exam
practise	8.1	practise exam
qualify	7.7	qualifying exam
administer	7.5	exam administered
retake	7.1	retake the exam
schedule	7.0	exam is scheduled

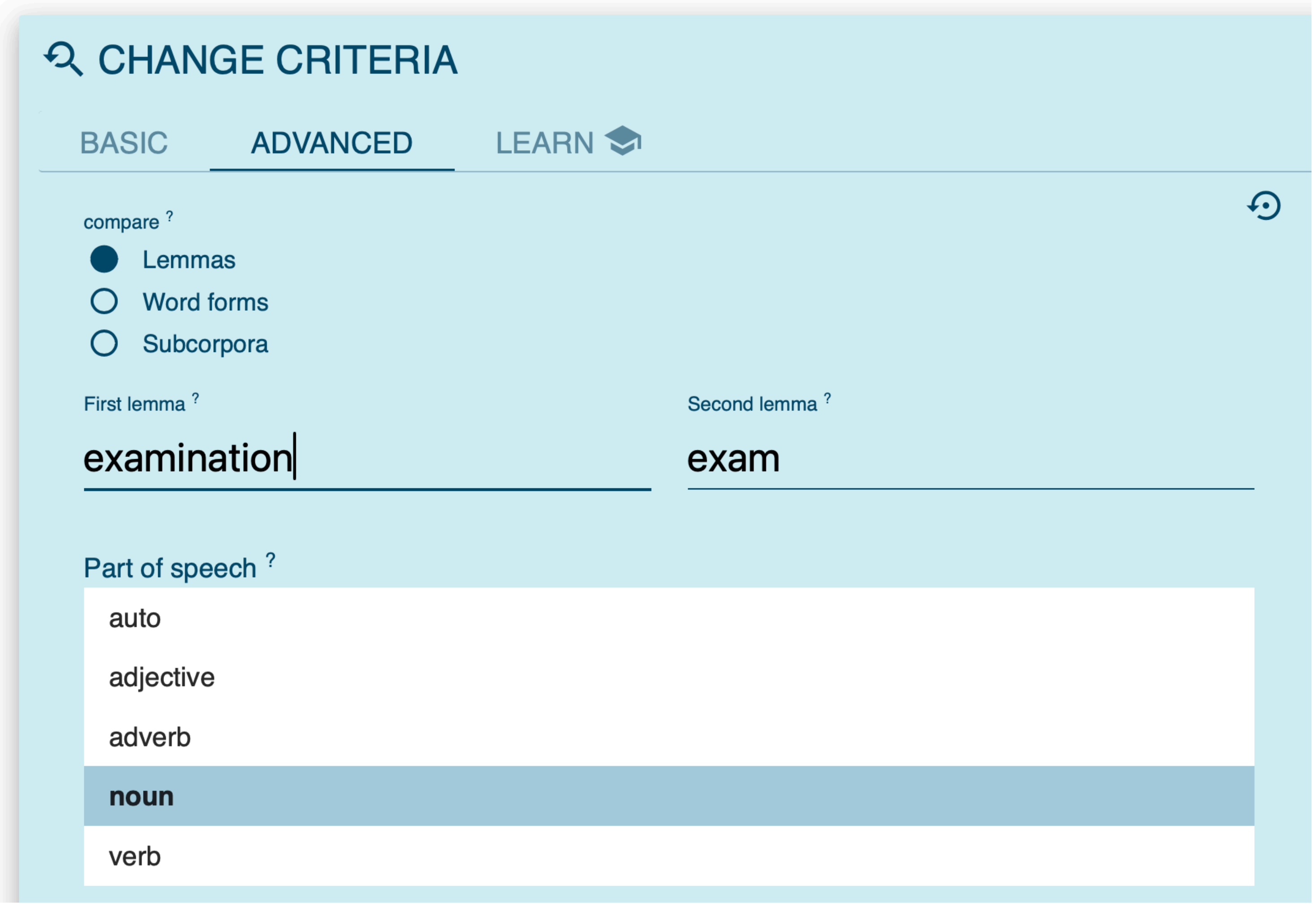
verbs with "exam" as subject		
dump	7.1	exam dumps
file	6.3	can get all exam files paying an astonishingly
test	5.8	exam tests
assess	5.7	exam assesses
consist	5.4	exam consists of
require	5.2	Exam requires the candidates to
loom	5.2	exams looming
grade	5.2	exam grading

3. Visualise results.



Word sketches: Comparison

1. Compare source words and clipped forms (e.g. examination vs exam).



2. Review shared and unique collocates.

examination 6,046x | exam 1,559x

adjective predicates of "examination/exam" | ... is a "examination/exam" | verbs with "examination/exam" as subject

"examination/exam" and/or ...		
assessment	42	0
examination	20	0
curriculum	16	0
dissertation	7	0
biopsy	7	0
course	12	3
test	26	8
essay	5	3
er	0	3
work	0	3
something	0	4
exam	0	8

verbs with "examination/exam" as object		
undergo	28	0
resit	9	0
conduct	28	0
perform	23	0
finish	6	4
take	89	74
sit	43	35
do	23	31
fail	20	20
pass	83	128
ride	0	3
flunk	0	3

modifiers of "examination/exam"		
post-mortem	60	0
histological	58	0
detailed	130	0
close	147	0
mortem	35	0
medical	125	4
entrance	35	18
matriculation	9	6
GCSE	10	9
scholarship	6	7
11-plus	0	4
maths	0	7

3. Inspect collocations in detail.

CONCORDANCE

British National Corpus (BNC)

Get more space

CQL er + exam • 3

0.03 per million tokens • 0.0000027%

KWIC

	Details	Left context	KWIC	Right context
1	Spoken context-...	ow is that you will need it for your **er** , your Q M	**exam**	not er your exam, your project right, cos there is
2	Spoken context-...	ould I like to do it and so of course I had to pass	**exams**	and **er** actually, can I read some notes that</s><
3	Spoken context-...	ell of course er I kn I know about take two more	**exams**	and **er** I know the second one was mental arithr

Guiding questions

1. What is the general semantic signature of the source word?
2. What is the general semantic signature of the clipped form?
3. In what ways do they differ (stylistic, social, formality)?
4. Do they occur in different syntagmatic contexts despite similar denotation?
5. Does the clipped form have a wider or narrower scope of meaning?

Summary

- **A difference in form (often) implies a difference in meaning.** (Principle of No Synonymy)
- **Meaning is use.** The contexts a word appears in tell us about its function. (Distributional Hypothesis)
- **Clippings are not just lazy shortcuts.** They develop unique semantic, pragmatic, and social functions, often signalling familiarity and appearing in more “involved” contexts.

References

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